

OOP2 →

Agenda

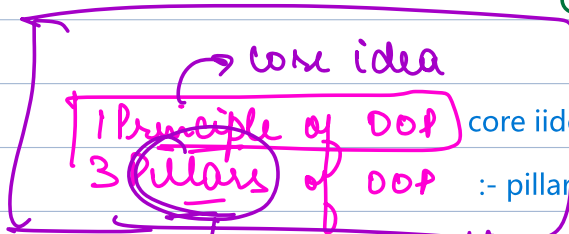
- ① Abstraction] ⇒
 - ② Encapsulation] ⇒
 - ③ Classes, Objects, etc
 - ④ Access Modifiers ⇒
 - ⑤ Constructors
- Default
→ Manual
→ Copy

Today

OOP Pillars of OOP

↳ Core concepts of OOP

↳ core feature / ideas of OOP



core idea why OOps should exist

:- pillars means how to do that

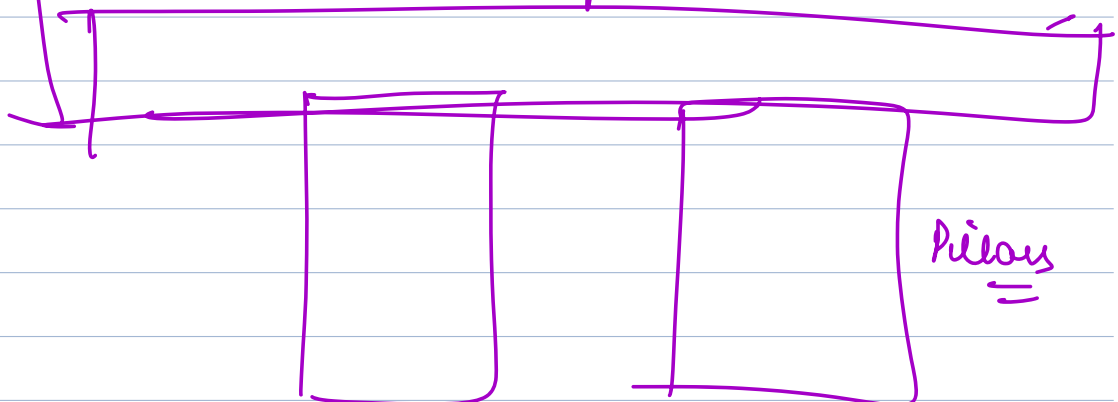
supports

support the idea

of OOP → how to bring OOP into practice

6/4/5/3/

4 Pillars of OOP



finally: object oriented programming is based on principle of abstraction which achieved by those three pillars

1 Principle of OOP $\Rightarrow$

Abstraction

to achieve this one principle which is abstraction we used the those three pillars

3 Pillars of OOP

$\Rightarrow$ Inheritance
Polymorphism
Encapsulation

existing as an idea (concept)
① ABSTRACTION

represent a software system in terms of ideas.

$\Rightarrow$ whatever S/W system you are implementing there will be many ideas in that

(Entities | anything about which you have to store info | Any thing that can cause)

represent that S/W in terms of ideas

Scaler's S/W system

$\Rightarrow$ Student

Assignment

$\Downarrow$
TA

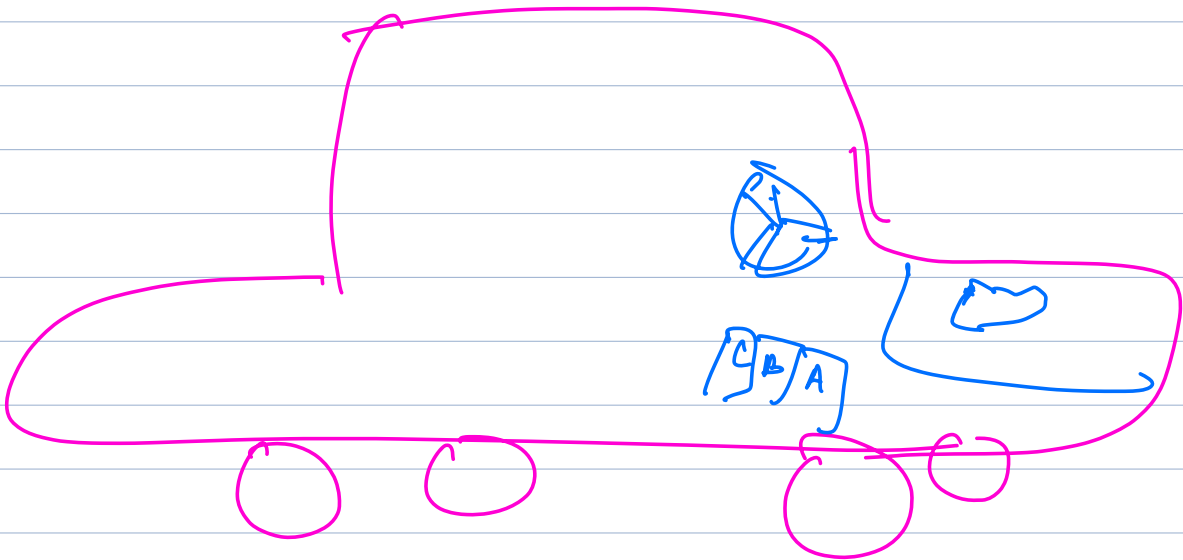
$\uparrow$
Course

Batch

Instructor

Note

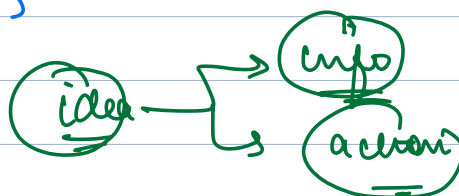
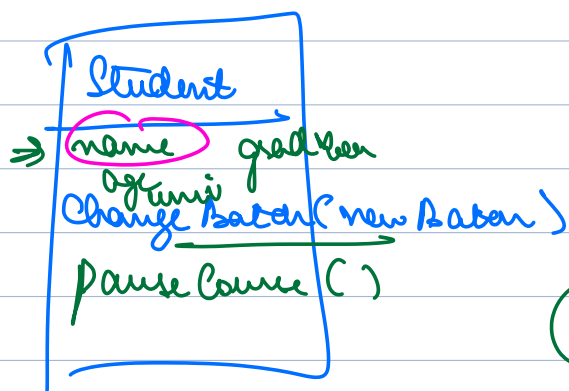
Class.



Car

Steering Wheel
→ move left
→ move right

Engine
capacity
make power
exp



Abstraction

- ① Rep a complex S/W system in terms of ideas:
 - a) Anything about which you are storing info. ✓
 - b) Anything on which you can perform actions. ✓
- ② Not everyone needs to know details of working of these ideas.

3 Pillars ⇒ Help us represent complex S/W system in terms of ideas.

① Encapsulation

Purpose of Encapsulation

- ① Hold info and actions of an idea together.
- ② Protect the info and actions against illegitimate access.

⇒ Holds the medicine together.

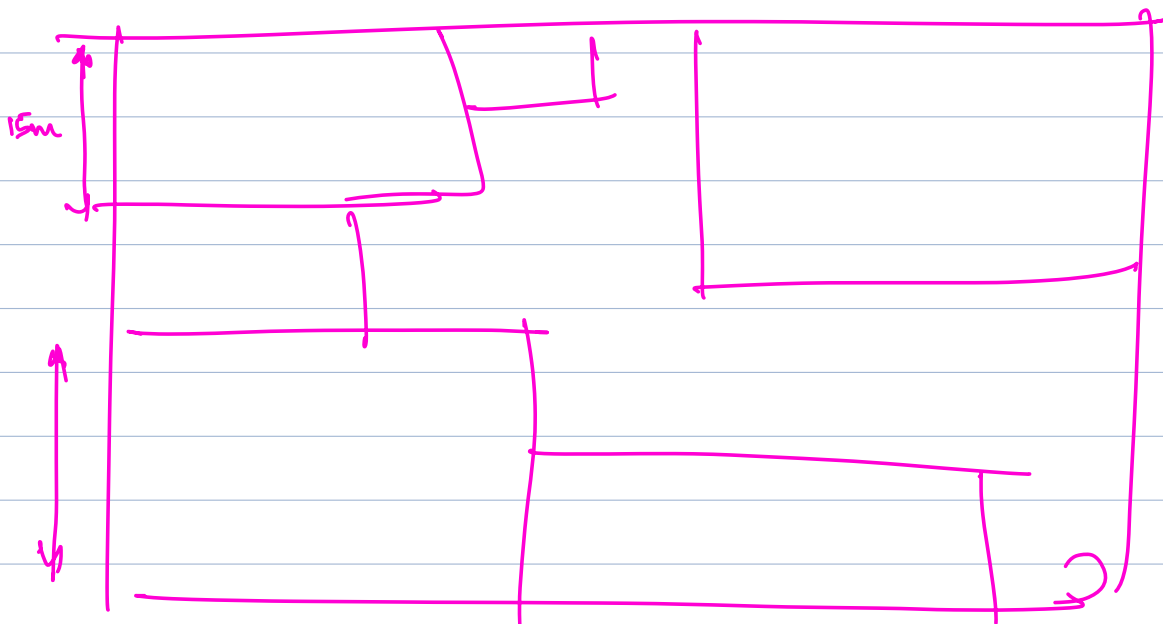
⇒ Protect the medicine from external environment.

Student!

Class

Batch

Class $\Rightarrow$ Blueprint of an idea



$\Rightarrow$ design of how every house formed via it will look like / what all will each a house have.

Blueprint

①

1 Blueprint $\Rightarrow$ Multiple Houses

②

Doesn't occupy any land

class

name of class

→ idea. (info and action)
→ what, every instance of that class should have

class Student {

String name;
String email;
String batch;
double psp;

void changeBatch (String newBatch) {
 batch = newBatch;
}

void applyForJob (int jobId) {
 if (psp >= 80) {
 print ("success")
 }
 print ("failure")
}

}

→ From one class you can create multiple instances.
→ Class doesn't take RAM space. Instance of class does.

OBJECT

- Instance of a class.
- occupy memory
- each object is totally isolated from other objects of that class.

